

Mewtant

Building Generative AI Image Application on AWS, Mewtant Attracts 1 Million Monthly Active Users

Using AWS, Mewtant built its AI image generator, Pixai.Art, within 3 months; reduced costs by 50 percent; gained scalability; and gained 1 million monthly active users.

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3 months

3 months from conception to launch



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



100 TB

100 TB of AI-generated image content



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



1 million

1 million monthly active users worldwide



Overview

The Pixai.Art artificial intelligence (AI) image generator launched by [Mewtant Inc.](#) (Mewtant) helps animation lovers to create a variety of anime images with text prompts and share them with community users. It has gained 1 million monthly active users worldwide.

Pixai.Art is built on Amazon Web Services (AWS) to accelerate inference performance, using model training provided by [Amazon SageMaker](#) — a managed service designed to build, train, and deploy machine learning models for any use case — and [Amazon Elastic Compute Cloud](#) (Amazon EC2) [Inf2 instances](#), which are purpose built for deep learning inference, supported by [Amazon Inferentia2](#), which provides the highest performance at the lowest cost in Amazon EC2. Mewtant released the first version of Pixai.Art within 3 months and brought more new experiences to anime enthusiasts and art creators through continual, rapid iteration.



Opportunity | Providing Image Generation and Sharing Services for Anime Enthusiasts

The emergence of generative AI provides more tech assistance for artistic creation and brings more convenience. Mewtant's goal is to provide users with an excellent virtual character generation experience. Its long-term vision is to surpass image generation capabilities and develop its own image and video models to explore the boundaries of visual narratives and create a brand-new story experience for users through multiple modalities.

As a startup team, Mewtant wants to quickly seize business opportunities and create business value in generative AI. "This is challenging for us," says Raven Gao, CEO of Mewtant. "We need to collaborate with cloud service providers with rich capabilities to complete construction and release in a cost-effective manner in a short time."



Solution | Building Pixai.Art on AWS to Unleash Users' Artistic Potential

When conceptualizing the Pixai.Art AI image generator, Mewtant hoped to create a solution where users could seamlessly transform their imaginations into charming artwork for free. Therefore, choosing a suitable cloud service provider and controlling costs was crucial. With these goals in mind, Mewtant decided to build Pixai.Art on AWS. Mewtant believed that the extensive use cases, rich services, and professional generative AI solutions of AWS could quickly turn their ideas into reality. “We knew that after Pixai.Art went live, it would face global users,” says Gao. “We felt that the extensive investment of AWS in infrastructure, good network, simple integration of various services, and support from professional teams would help us develop our business in generative AI.”

With the support of the AWS Architect team, Mewtant first used a series of pretrained models built into Amazon SageMaker and [Amazon SageMaker JumpStart](#) — an ML hub with foundation models, built-in algorithms, and prebuilt ML solutions — for proof of concept of AI image generators, which achieved the team’s expected results. This was a stable foundation for the construction of the entire application and helped Mewtant accumulate experience in cloud-based ML and AI application construction. “The convenience of Amazon SageMaker exceeded our expectations, and some built-in, pretrained, text-generated image models in Amazon SageMaker JumpStart were simple to use,” says Alvin Li, a senior technical staff at Mewtant. “After testing, we began deploying our own models in Amazon SageMaker, which were optimized and fine-tuned for anime and used for Pixai.Art construction. We have achieved satisfactory results.” To accelerate the inference process of generative AI, Mewtant chose the Amazon EC2 Inf2 instances supported by Amazon Inferentia2 because of their suitability for generative AI scenarios and ability to provide Mewtant with the required inference performance.

Mewtant planned to synchronously launch Pixai.Art on both the web and mobile applications. After deploying a suitable text-generated image model, the team needed to build stable support for both the web and mobile applications. Mewtant adopted [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed Kubernetes service, to provide an automatically scalable cluster for applications. Through this method, the backend of the application can adjust resources as needed based on the number of users and the size of the workload, providing stable resource support for Pixai.Art’s initial operation and long-term user growth. The Mewtant team also chose [Amazon Relational Database Service](#) (Amazon RDS) — a collection of managed services that makes it simple to operate, and scale databases in the cloud — to host relational databases for Pixai.Art user data. Amazon RDS, with its rich database management experience, similar to that of open-source database products, helped Mewtant to simplify database management, and the service’s high availability and security protects user data.



Image storage and management is also important for Pixai.Art, so Mewtant uses [Amazon Simple Storage Service](#) (Amazon S3) — an object storage service offering cutting-edge scalability, data availability, security, and performance — to store various types of anime images generated by users. To automatically generate image thumbnails and adjust resolution using serverless services and event-triggered code, Mewtant also uses [AWS Lambda](#), a serverless, event-driven compute service. With these two services and [Amazon CloudFront](#) — a content delivery network service built for high performance, security, and developer convenience — the generated images are quickly displayed in users' browsers or mobile applications.

After the official launch of Pixai.Art, users around the world began to showcase their artistic creations with this powerful generative AI and image editing tool. With just a few simple steps, images can be created and shared with community users, stimulating more artistic potential and creativity while enjoying the fun.

Outcome | Launching Pixai.Art within 3 Months and Shortening Functional Iterations to 1–2 Days

Using AWS, Mewtant released Pixai.Art on the web and mobile devices in just 3 months, successfully attracting the first batch of users interested in anime culture and generative AI and quickly accumulating 1 million monthly active users. "The efficient integration of Amazon SageMaker and a series of AWS services and functions helps us to focus more on innovation at the functional level without worrying about the configuration and management of underlying services," says Li. "Now, we can shorten the iteration of a single function to 1–2 days, which also means that our users can come up with optimization ideas and gain new experiences at any time."

As the number of active users increases, business costs are a key concern for Mewtant. To optimize infrastructure costs, Mewtant adopted Amazon EC2 Spot Instances, which let you take advantage of unused Amazon EC2 capacity in the AWS Cloud, saving the team 50 percent in compute costs. In terms of operation and maintenance, due to the extensive application of managed services in Pixai.Art, the team only needs to arrange one technical personnel to achieve the operation and maintenance management of the entire solution, helping Mewtant to invest more technical strength in the research and development of generative AI.

Pixai.Art also benefits from the scale of AWS and the elastic scalability of AWS services, which have helped Mewtant keep up with the continual growth of the amount of data and the number of users. "At present, we maintain hundreds of millions of user-related data points for our applications, and Amazon RDS does not require us to put in extra efforts in terms of computing power expansion or storage expansion," says Li. "In terms of image storage and management, Amazon S3 helps us to store over 100 TB of images at an appropriate cost and integrate with Amazon CloudFront to achieve rapid content delivery, which provides users with a faster image acquisition experience. In the past 6 months, as the number of users has grown by 15 times, AWS has given us confidence to keep our business running smoothly."

In the future, Mewtant will continue to deepen the development of generative AI application scenarios based on Pixai.Art AI image generators and will continue to integrate innovative services from AWS into solutions for multimodal applications, such as images, videos, and storytelling. It will use hosted services such as [Amazon Bedrock](#), a fully managed service that offers a choice of high-performing foundation models from leading AI companies, to improve inference speed, and reduce costs to provide users with more interesting generative AI experiences.

Mewtant was founded by a group of anime and virtual world enthusiasts who were aiming to provide a great virtual character generation experience. Headquartered in Singapore, it is a global team with members working in Singapore, the United States, Canada, and Japan. With Pixai.Art's help, users can generate virtual characters in seconds by simply typing in a few lines.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more>>](#)

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

[Learn more>>](#)

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more>>](#)

Amazon CloudFront

Amazon CloudFront is a content delivery network (CDN) service built for high performance, security, and developer convenience.



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